

Inequalities — Solutions

Problem 1. Let n be a positive integer, and let $x_1, x_2, \dots, x_n$ be positive real numbers such that $x_j \geq 1$ for any j . Prove that

$$1 + x_1 x_2 \cdots x_n \geq \frac{(1 + x_1)(1 + x_2) \cdots (1 + x_n)}{2^{n-1}}.$$

Solution. We prove the inequality by induction on n . The base case $n = 1$ is trivial since the two sides are equal. Assume

$$1 + x_1 x_2 \cdots x_k \geq \frac{(1 + x_1)(1 + x_2) \cdots (1 + x_k)}{2^{k-1}}. \quad (1)$$

for some positive integer k . We need to prove

$$1 + x_1 x_2 \cdots x_{k+1} \geq \frac{(1 + x_1)(1 + x_2) \cdots (1 + x_{k+1})}{2^k}. \quad (2)$$

By (1), we have

$$\frac{(1 + x_1)(1 + x_2) \cdots (1 + x_{k+1})}{2^k} \leq (1 + x_1 x_2 \cdots x_k) \frac{1 + x_{k+1}}{2}. \quad (3)$$

Let $a = x_1 x_2 \cdots x_k$ and $b = x_{k+1}$. Comparing (2) and (3), it suffices to prove

$$\frac{(1 + a)(1 + b)}{2} \leq 1 + ab.$$

This is equivalent to $(a - 1)(b - 1) \geq 0$. This is true since $a = x_1 x_2 \cdots x_k \geq 1$ and $b = x_{k+1} \geq 1$. Thus, the inequality is proven by induction. Equality holds when $n - 1$ of $x_1, x_2, \dots, x_n$ are equal to 1. $\square$

Problem 2. (Vietnam MO 2011 Problem 1) Prove that if $x > 0$ and $n \in \mathbb{N}$, then we have

$$\frac{x^n(x^{n+1} + 1)}{x^n + 1} \leq \left(\frac{x + 1}{2}\right)^{2n+1}.$$

Solution. We prove the result by induction on $n \geq 0$. The base case $n = 0$ is trivial as the two sides are equal. Assume

$$\frac{x^k(x^{k+1} + 1)}{x^k + 1} \leq \left(\frac{x + 1}{2}\right)^{2k+1} \quad (1)$$

for some $k \geq 0$. We need to prove

$$\frac{x^{k+1}(x^{k+2} + 1)}{x^{k+1} + 1} \leq \left(\frac{x + 1}{2}\right)^{2k+3}.$$

By (1), we have

$$\left(\frac{x+1}{2}\right)^{2k+3} \geq \frac{x^k(x^{k+1}+1)}{x^k+1} \left(\frac{x+1}{2}\right)^2.$$

It suffices to prove

$$\frac{x^k(x^{k+1}+1)}{x^k+1} \left(\frac{x+1}{2}\right)^2 \geq \frac{x^{k+1}(x^{k+2}+1)}{x^{k+1}+1},$$

which means

$$(x^{k+1}+1)^2(x+1)^2 \geq 4x(x^{k+2}+1)(x^k+1).$$

After expansion and factorization, this is exactly

$$(x^{k+1}-1)^2(x-1)^2 \geq 0,$$

which is obviously true. This proves the inequality by induction. Equality holds when $x = 1$. $\square$

Problem 3. (Kazakhstan MO 2012 Grade 10 Problem 1) Let $x_1, x_2, \dots, x_n$ be positive real numbers. Prove the inequality

$$\frac{1}{x_1+1} + \frac{1}{x_2+1} + \dots + \frac{1}{x_n+1} \leq \frac{n}{1 + \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}}.$$

Solution. Let $a_k = \frac{1}{x_k}$ for $k = 1, 2, \dots, n$. We have to prove

$$\frac{a_1}{a_1+1} + \frac{a_2}{a_2+1} + \dots + \frac{a_n}{a_n+1} \leq \frac{n}{1 + \frac{n}{a_1+a_2+\dots+a_n}}.$$

Let $f(t) = \frac{t}{t+1} = 1 - \frac{1}{t+1}$. Then we have $f'(t) = \frac{1}{(t+1)^2}$ and $f''(t) = -\frac{2}{(t+1)^3} < 0$ for $t > 0$. This shows f is concave. By Jensen's inequality, we have

$$f(a_1) + f(a_2) + \dots + f(a_n) \leq nf\left(\frac{a_1+a_2+\dots+a_n}{n}\right) = \frac{n}{1 + \frac{n}{a_1+a_2+\dots+a_n}}.$$

This proves the desired inequality. Equality holds when $x_1 = x_2 = \dots = x_n$. $\square$

Problem 4. Let n be a positive integer, and let $x_1, x_2, \dots, x_n$ be positive real numbers satisfying $x_1 x_2 \dots x_n = 1$. Prove that there exists $k \in \{1, 2, \dots, n\}$ such that

$$\frac{\sqrt[k]{x_k} + \sqrt[k+1]{x_{k+1}} + \dots + \sqrt[n]{x_n}}{n+1-k} \geq 1.$$

Solution. Suppose on the contrary that no such k exists. This means

$$\sqrt[k]{x_k} + \sqrt[k+1]{x_{k+1}} + \cdots + \sqrt[n]{x_n} < n + 1 - k$$

for each $k = 1, 2, \dots, n$. Summing up all these inequalities, we obtain

$$x_1 + 2\sqrt{x_2} + 3\sqrt[3]{x_3} + \cdots + n\sqrt[n]{x_n} < \frac{n(n+1)}{2}.$$

However, by the AM-GM inequality, we have

$$x_1 + 2\sqrt{x_2} + 3\sqrt[3]{x_3} + \cdots + n\sqrt[n]{x_n} \geq (1 + 2 + \cdots + n)(x_1 x_2 \cdots x_n)^{\frac{1}{1+2+\cdots+n}} = \frac{n(n+1)}{2}.$$

This is a contradiction. □

Problem 5. Let a, b, c, d be positive real numbers satisfying $a + b + c + d = 4$. Prove that

$$\frac{a}{b(b+1)} + \frac{b}{c(c+1)} + \frac{c}{d(d+1)} + \frac{d}{a(a+1)} \geq \frac{8}{(a+c)(b+d)}.$$

Solution. Define the function $f(x) = \frac{1}{x(x+1)} = \frac{1}{x} - \frac{1}{x+1}$. We have $f'(x) = -\frac{1}{x^2} + \frac{1}{(x+1)^2}$ and $f''(x) = \frac{2}{x^3} - \frac{2}{(x+1)^3} > 0$ for any $x > 0$. Therefore, f is a convex function. By Jensen's inequality, we have

$$\frac{af(b) + bf(c) + cf(d) + df(a)}{a + b + c + d} \geq f\left(\frac{ab + bc + cd + da}{a + b + c + d}\right).$$

Let $t = ab + bc + cd + da = (a+c)(b+d)$. Then we have

$$af(b) + bf(c) + cf(d) + df(a) \geq 4f\left(\frac{t}{4}\right) = \frac{64}{t(t+4)}.$$

We only need to prove $\frac{64}{t(t+4)} \geq \frac{8}{t}$. This is equivalent to $t \leq 4$, which holds since

$$t = (a+c)(b+d) \leq \left(\frac{(a+c) + (b+d)}{2}\right)^2 = 4$$

by the AM-GM inequality. This completes the proof. Equality holds when $a = b = c = d = 1$. □

Problem 6. (USAMO 1980 Problem 5) Prove that for numbers a, b, c in the interval $[0, 1]$,

$$\frac{a}{b+c+1} + \frac{b}{c+a+1} + \frac{c}{a+b+1} + (1-a)(1-b)(1-c) \leq 1.$$

Solution. Let

$$f(a, b, c) = \frac{a}{b+c+1} + \frac{b}{c+a+1} + \frac{c}{a+b+1} + (1-a)(1-b)(1-c).$$

For fixed $b, c \in [0, 1]$, define $g(a) = f(a, b, c)$. We see that

$$\begin{aligned} g'(a) &= \frac{1}{b+c+1} - \frac{b}{(c+a+1)^2} - \frac{c}{(a+b+1)^2} - (1-b)(1-c), \\ g''(a) &= \frac{2b}{(c+a+1)^3} + \frac{2c}{(a+b+1)^3} \geq 0. \end{aligned}$$

This shows g is convex. Therefore, the maximum value of g is attained at the endpoint $a = 0$ or $a = 1$. In each case $a = 0$ or $a = 1$, we consider $h(b) = f(a, b, c)$. The same arguments show that h attains its maximum value at $b = 0$ or $b = 1$. Similarly, in each subcase, the maximum value of $f(a, b, c)$ is attained at $c = 0$ or $c = 1$. Thus, it suffices to consider $a, b, c \in \{0, 1\}$. We easily find that

$$f(0, 0, 0) = f(0, 0, 1) = f(0, 1, 1) = f(1, 1, 1) = 1.$$

Therefore, the maximum value of $f(a, b, c)$ is 1. This proves the inequality. Equality holds when $a, b, c \in \{0, 1\}$. $\square$

Problem 7. (Japan MO 2002 Problem 4) Let $n \geq 3$ be a natural number, and let $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$ be positive real numbers such that $a_1 + a_2 + \dots + a_n = 1$ and $b_1^2 + b_2^2 + \dots + b_n^2 = 1$. Prove that

$$a_1(b_1 + a_2) + a_2(b_2 + a_3) + \dots + a_n(b_n + a_1) < 1.$$

Solution. By the Cauchy-Schwarz inequality, we have

$$\begin{aligned} & a_1(b_1 + a_2) + a_2(b_2 + a_3) + \dots + a_n(b_n + a_1) \\ &= (a_1b_1 + a_2b_2 + \dots + a_nb_n) + (a_1a_2 + a_2a_3 + \dots + a_na_1) \\ &\leq \sqrt{(a_1^2 + a_2^2 + \dots + a_n^2)(b_1^2 + b_2^2 + \dots + b_n^2)} + (a_1a_2 + a_2a_3 + \dots + a_na_1) \\ &\leq \sqrt{a_1^2 + a_2^2 + \dots + a_n^2} + \frac{1}{2}((a_1 + a_2 + \dots + a_n)^2 - (a_1^2 + a_2^2 + \dots + a_n^2)) \\ &= \sqrt{x} + \frac{1-x}{2} \end{aligned}$$

where $x = a_1^2 + a_2^2 + \dots + a_n^2$. We now show that this is at most 1. Indeed,

$$\begin{aligned} & \sqrt{x} + \frac{1-x}{2} \leq 1 \\ \Leftrightarrow & \sqrt{x} \leq \frac{x+1}{2} \\ \Leftrightarrow & x \leq \frac{(x+1)^2}{4} \\ \Leftrightarrow & 0 \leq (x-1)^2. \end{aligned}$$

This is obvious. Equality holds when $x = 1$ and $n = 3$. However, since $f(t) = t^2$ is convex, and $(1, 0, 0, \dots, 0)$ majorizes $(a_1, a_2, \dots, a_n)$ (assuming $a_1 \geq a_2 \geq \dots \geq a_n$), we have

$$x = f(a_1) + f(a_2) + \dots + f(a_n) < f(1) + f(0) + f(0) + \dots + f(0) = 1.$$

Therefore, equality cannot hold, and the inequality is proven. $\square$

Problem 8. (Brazil MO 2011 Problem 6) Let $a_1, a_2, a_3, \dots, a_{2011}$ be nonnegative reals with sum $\frac{2011}{2}$. Prove that

$$|(a_1 - a_2)(a_2 - a_3) \cdots (a_{2011} - a_1)| \leq \frac{3\sqrt{3}}{16}.$$

Solution. Let $f = |(a_1 - a_2)(a_2 - a_3) \cdots (a_{2011} - a_1)|$. We refine the sequence $\{a_n\}$ (where we define $a_{n+2011} = a_n$) as follows to maximize f .

- (i) If there exist 3 consecutive terms a, b, c such that $a \geq b > 0$ and $b \leq c$, we can replace b by 0 and the largest term M in the sequence by $M + b$. Since $(a - b)(c - b) < (a - 0)(c - 0)$ and $(M - x)(M - y) < (M + b - x)(M + b - y)$ for any $x, y \leq M$, this increases the value of f .
- (ii) If there exist 4 consecutive terms $0, a, b, c$ (or $c, b, a, 0$) such that $0 < a \leq b \leq c$, we can swap a and b . Since $(a - 0)(b - a)(c - b) \leq (b - 0)(b - a)(c - a)$, the value of f is increased. Then we have 3 consecutive terms b, a, c with $b \geq a > 0$ and $a \leq c$. So we can apply (i) to replace a by 0.
- (iii) If there exist 5 consecutive terms $0, a, b, c, 0$ such that $0 < a \leq b$ and $b \geq c > 0$, we may assume $a \geq c$ (otherwise we swap a and c without changing f). We replace b and c by 0 and $b + c$ respectively. As $(b - a)(b - c)c \leq (b - c)^2 c < c(b + c)^2 \leq a(b + c)^2$, the value of f is increased.

Note that one more term becomes 0 after each refinement. Therefore, eventually no more refinement can be carried out. At this moment at least one term is 0, since otherwise we can apply (i) with b being the smallest term. Consider some consecutive terms $0, b_1, b_2, \dots, b_k, 0$ where $b_j \neq 0$ for each j . If $k = 0$, then $f = 0$, which is clearly not the maximum value. So we may assume $k \geq 1$.

We claim that k must be 1 or 2. Suppose on the contrary that $k \geq 3$. Firstly, we have $0 < b_1$. If $b_1 \geq b_2$, then $b_2 > b_3$ or otherwise we can apply (i). Similarly, we have $b_3 > b_4 > \dots > b_k > 0$. But then we can apply (ii) since $k \geq 3$, contradiction. Thus, we must have $0 < b_1 < b_2$. If $b_2 \leq b_3$, we can apply (ii). So we may assume $b_2 > b_3$. As above, we have $b_2 > b_3 > \dots > b_k > 0$. If $k \geq 4$, we can apply (ii). But if $k = 3$, we can apply (iii). We get a contradiction in any case.

Thus, there is exactly 1 or 2 positive terms between any two consecutive 0's. For the consecutive terms $0, x, 0$, the contribution of the differences of these terms to f is x^2 . For the consecutive terms

$0, y, z, 0$ (WLOG assume $y \leq z$), the contribution of the differences of these terms to f is $yz(z - y)$, and we have

$$\begin{aligned} yz(z - y) &= \frac{1}{2}((\sqrt{3} + 1)y)((\sqrt{3} - 1)z)(z - y) \\ &\leq \frac{1}{2} \left(\frac{(\sqrt{3} + 1)y + (\sqrt{3} - 1)z + (z - y)}{3} \right)^3 \\ &= \frac{\sqrt{3}(y + z)^3}{18} \end{aligned}$$

by the AM-GM inequality. Equality holds when $(\sqrt{3} + 1)y = (\sqrt{3} - 1)z = z - y$. This means that $y : z = (\sqrt{3} - 1) : (\sqrt{3} + 1)$.

Let $x_1, x_2, \dots, x_m$ be the terms adjacent to two 0's, and let $y_1, y_2, \dots, y_n$ be the sum of the two terms lying between two consecutive 0's. Note that $2m + 3n = 2011$. Let $X = x_1 + x_2 + \dots + x_m$ and $Y = y_1 + y_2 + \dots + y_n$ so that $X + Y = \frac{2011}{2}$. From above and by the AM-GM inequality, we have

$$\begin{aligned} f &\leq x_1^2 x_2^2 \dots x_m^2 \left(\frac{\sqrt{3}}{18} \right)^n y_1^3 y_2^3 \dots y_n^3 \\ &\leq \left(\frac{X}{m} \right)^{2m} \left(\frac{\sqrt{3}}{18} \right)^n \left(\frac{Y}{n} \right)^{3n} \\ &= 2^{2m} \left(\frac{3\sqrt{3}}{2} \right)^n \left(\frac{X}{2m} \right)^{2m} \left(\frac{Y}{3n} \right)^{3n} \\ &\leq 2^{2m} \left(\frac{3\sqrt{3}}{2} \right)^n \left(\frac{X + Y}{2m + 3n} \right)^{2m + 3n} \\ &= 2^{2011 - 3n} \left(\frac{3\sqrt{3}}{2} \right)^n \frac{1}{2^{2011}} \\ &= \left(\frac{3\sqrt{3}}{16} \right)^n \\ &\leq \frac{3\sqrt{3}}{16} \end{aligned}$$

since $\frac{3\sqrt{3}}{16} < 1$ and $n \geq 1$ (as $2m + 3n = 2011$, we cannot have $n = 0$). This proves the inequality.

Equality holds when the sequence is of the form $0, 1, 0, 1, \dots, 0, 1, 0, \frac{3 - \sqrt{3}}{4}, \frac{3 + \sqrt{3}}{4}$. $\square$